

Dynamic programming

Instructor

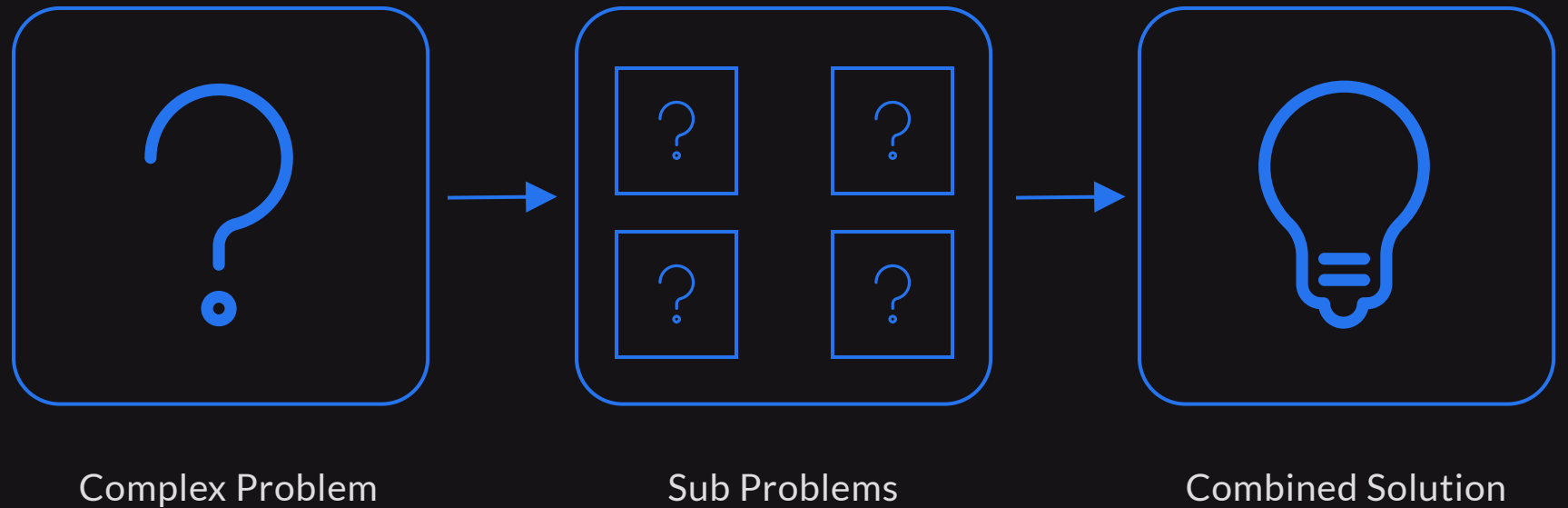
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Introduction to Dynamic Programming

Dynamic Programming involves breaking down a complex problem into subproblems, solving them and then combining the solutions



Dynamic Programming based on Principle of Optimality

Principle of Optimality



Principle of Optimality:

Subsolutions of an optimal solution of the problem are themselves optimal solutions for their subproblems

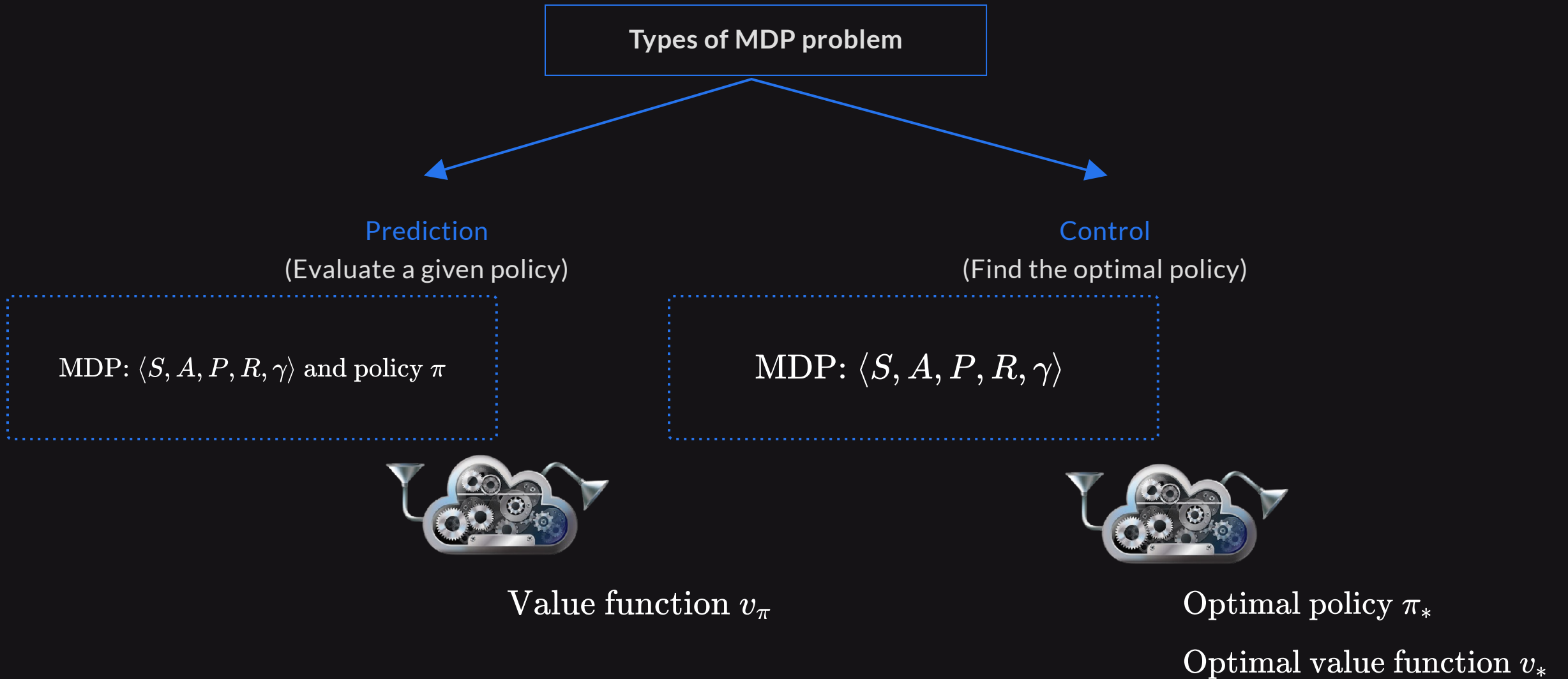
Why Dynamic Programming?

- Same subproblems can recur multiple times
- We can cache their solutions and reuse

Why Dynamic Programming?

- MDPs have this subproblem structure
 - Once you reach a state, future is independent of how you arrived at this state
 - One can store value functions for previously visited states and reuse them
- Dynamic programming assumes full knowledge of MDPs
- Can be used for both planning and control problems

Classification of MDP problem



Thank You
